

Aerodynamics: General Wing Relations

Ehsan Roohi

Ferdowsi University of Mashhad

GENERAL LIFT DISTRIBUTION

Consider the transformation

$$y = -\frac{b}{2} \cos \theta$$

where the coordinate in the spanwise direction is now given by θ , with $0 \leq \theta \leq \pi$. In terms of θ , the elliptic lift distribution given by Equation (5.31) is written as

$$\Gamma(\theta) = \Gamma_0 \sin \theta \quad \textbf{[5.47]}$$

Equation (5.47) hints that a Fourier sine series would be an appropriate expression for the general circulation distribution along an arbitrary finite wing. Hence, assume for the general case that

$$\Gamma(\theta) = 2bV_\infty \sum_1^N A_n \sin n\theta \quad \textbf{[5.48]}$$

where as many terms N in the series can be taken as we desire for accuracy. The coefficients A_n (where $n = 1, \dots, N$) in Equation (5.48) are unknowns; however, they must satisfy the fundamental equation of Prandtl's lifting-line theory; that is, the A_n 's must satisfy Equation (5.23). Differentiating Equation (5.48), we obtain

$$\frac{d\Gamma}{dy} = \frac{d\Gamma}{d\theta} \frac{d\theta}{dy} = 2bV_\infty \sum_1^N n A_n \cos n\theta \frac{d\theta}{dy} \quad \textbf{[5.49]}$$

Substituting Equations (5.48) and (5.49) into (5.23), we obtain

$$\alpha(\theta_0) = \frac{2b}{\pi c(\theta_0)} \sum_1^N A_n \sin n\theta_0 + \alpha_{L=0}(\theta_0) + \frac{1}{\pi} \int_0^\pi \frac{\sum_1^N n A_n \cos n\theta}{\cos \theta - \cos \theta_0} d\theta$$

The integral in Equation (5.50) is the standard form given by Equation (4.26). Hence, Equation (5.50) becomes

$$\alpha(\theta_0) = \frac{2b}{\pi c(\theta_0)} \sum_1^N A_n \sin n\theta_0 + \alpha_{L=0}(\theta_0) + \sum_1^N n A_n \frac{\sin n\theta_0}{\sin \theta_0} \quad \mathbf{[5.51]}$$

Examine Equation (5.51) closely. It is evaluated at a given spanwise location; hence, θ_0 is specified. In turn, b , $c(\theta_0)$, and $\alpha_{L=0}(\theta_0)$ are known quantities from the geometry and airfoil section of the finite wing. The only unknowns in Equation (5.51) are the A_n 's. Hence, written at a given spanwise location (a specified θ_0), Equation (5.51) is one algebraic equation with N unknowns, $A_1, A_2, \dots, A_n$. However, let us choose N different spanwise stations, and let us evaluate Equation (5.51) at each of these N stations. We then obtain a system of N independent algebraic equations with N unknowns, namely, $A_1, A_2, \dots, A_N$. In this fashion, actual numerical values are obtained for the A_n 's—numerical values that ensure that the general circulation distribution given by Equation (5.48) satisfies the fundamental equation of finite-wing theory, Equation (5.23).

Now that $\Gamma(\theta)$ is known via Equation (5.48), the lift coefficient for the finite wing follows immediately from the substitution of Equation (5.48) into (5.26):

$$C_L = \frac{2}{V_\infty S} \int_{-b/2}^{b/2} \Gamma(y) dy = \frac{2b^2}{S} \sum_1^N A_n \int_0^\pi \sin n\theta \sin \theta d\theta \quad \mathbf{[5.52]}$$

In Equation (5.52), the integral is

$$\int_0^\pi \sin n\theta \sin \theta d\theta = \begin{cases} \pi/2 & \text{for } n = 1 \\ 0 & \text{for } n \neq 1 \end{cases}$$

Hence, Equation (5.52) becomes

$$C_L = A_1 \pi \frac{b^2}{S} = A_1 \pi AR \quad \mathbf{[5.53]}$$

The induced drag coefficient is obtained from the substitution of Equation (5.48) into Equation (5.30) as follows:

$$\begin{aligned} C_{D,i} &= \frac{2}{V_\infty S} \int_{-b/2}^{b/2} \Gamma(y) \alpha_i(y) dy \\ &= \frac{2b^2}{S} \int_0^\pi \left(\sum_1^N A_n \sin n\theta \right) \alpha_i(\theta) \sin \theta d\theta \end{aligned} \quad \mathbf{[5.54]}$$

$$\begin{aligned}\alpha_i(y_0) &= \frac{1}{4\pi V_\infty} \int_{-b/2}^{b/2} \frac{(d\Gamma/dy) dy}{y_0 - y} \\ &= \frac{1}{\pi} \sum_1^N n A_n \int_0^\pi \frac{\cos n\theta}{\cos \theta - \cos \theta_0} d\theta\end{aligned}$$

The integral in Equation (5.55) is the standard form given by Equation (4.26). Hence, Equation (5.55) becomes

$$\alpha_i(\theta) = \sum_1^N n A_n \frac{\sin n\theta}{\sin \theta}$$

$$C_{D,i} = \frac{2b^2}{S} \int_0^\pi \left(\sum_1^N A_n \sin n\theta \right) \left(\sum_1^N n A_n \sin n\theta \right) d\theta \quad \int_0^\pi \sin m\theta \sin k\theta = \begin{cases} 0 & \text{for } m \neq k \\ \pi/2 & \text{for } m = k \end{cases}$$

$$C_{D,i} = \frac{2b^2}{S} \left(\sum_1^N n A_n^2 \right) \frac{\pi}{2} = \pi AR \sum_1^N n A_n^2$$

$$= \pi AR \left(A_1^2 + \sum_2^N n A_n^2 \right)$$

$$= \pi AR A_1^2 \left[1 + \sum_2^N n \left(\frac{A_n}{A_1} \right)^2 \right]$$

$$C_{D,i} = \frac{C_L^2}{\pi AR} (1 + \delta)$$

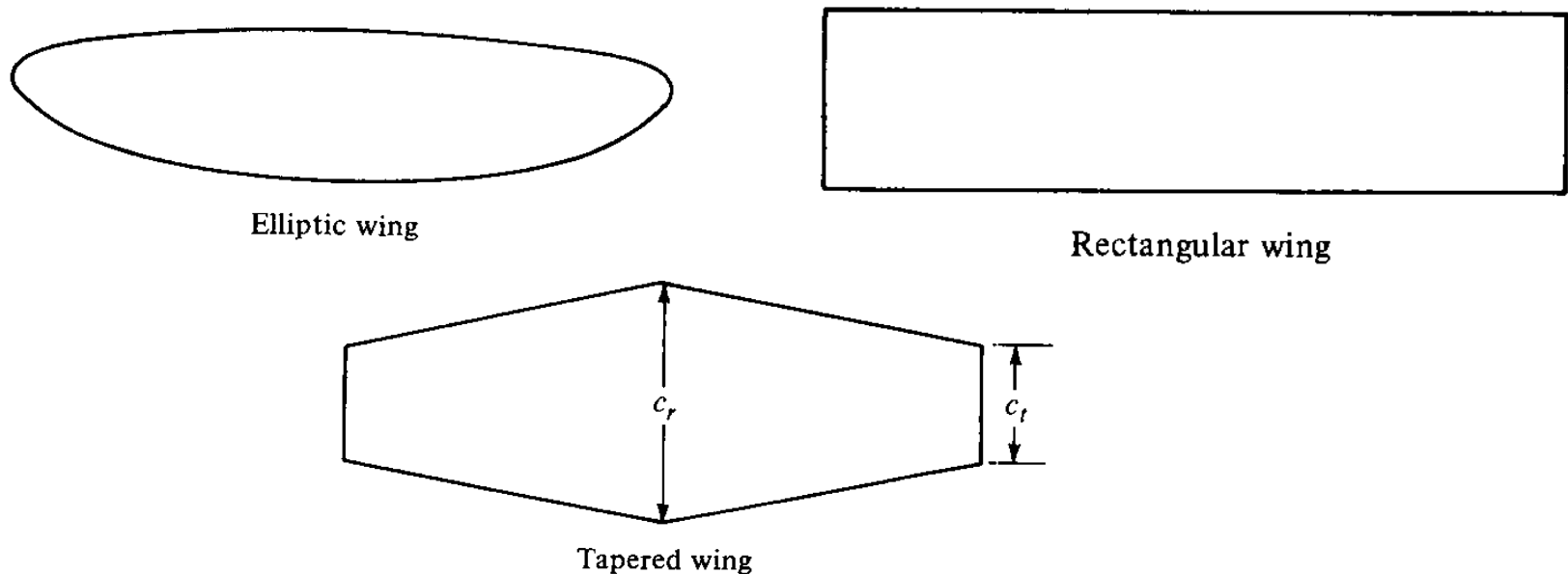
$$\delta = \sum_2^N n (A_n/A_1)^2$$

$$e = (1 + \delta)^{-1}$$

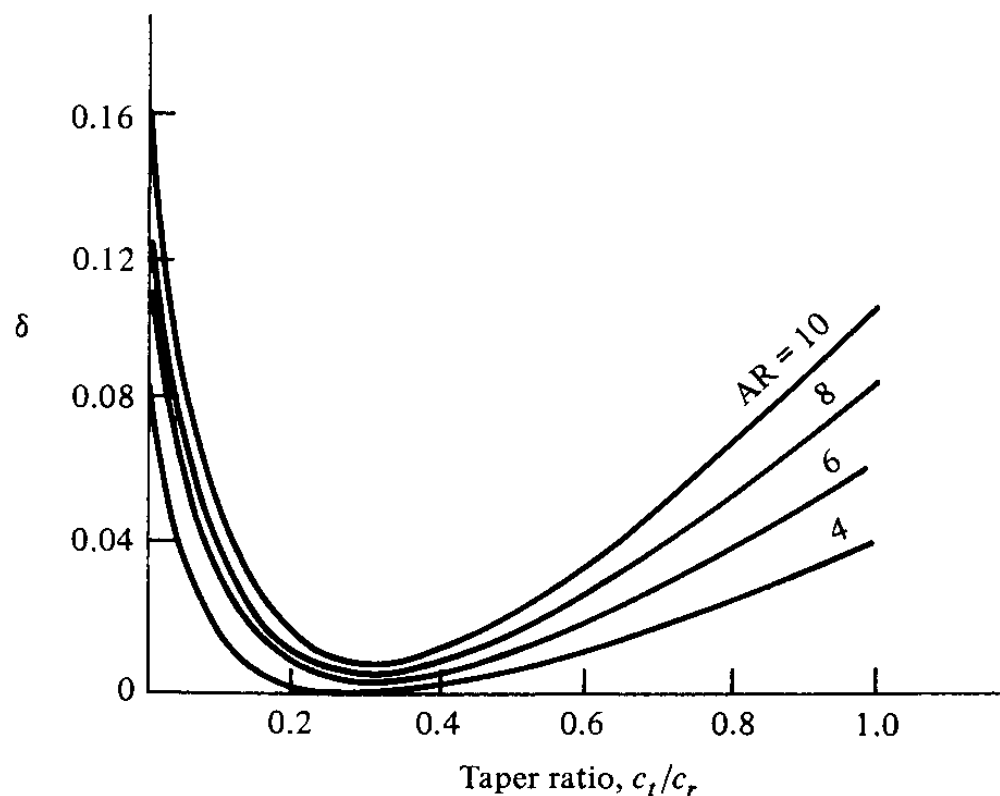
$$C_{D,i} = \frac{C_L^2}{\pi e AR}$$

for the elliptical lift distribution. Hence, the lift distribution which yields minimum induced drag is the *elliptical lift distribution*. This is why we have a practical interest in the elliptical lift distribution.

at the top of Figure 5.16. Several aircraft have been designed in the past with elliptical wings; the most famous, perhaps, being the British Spitfire from World War II, shown in Figure 5.17. However, elliptic planforms are more expensive to manufacture than, say, a simple rectangular wing as sketched in the middle of Figure 5.16. On the other hand, a rectangular wing generates a lift distribution far from optimum. A compromise is the tapered wing shown at the bottom of Figure 5.16. The tapered wing can be designed with a taper ratio, that is, tip chord/root chord $\equiv c_t/c_r$, such that the lift distribution closely approximates the elliptic case. The variation of δ as a function

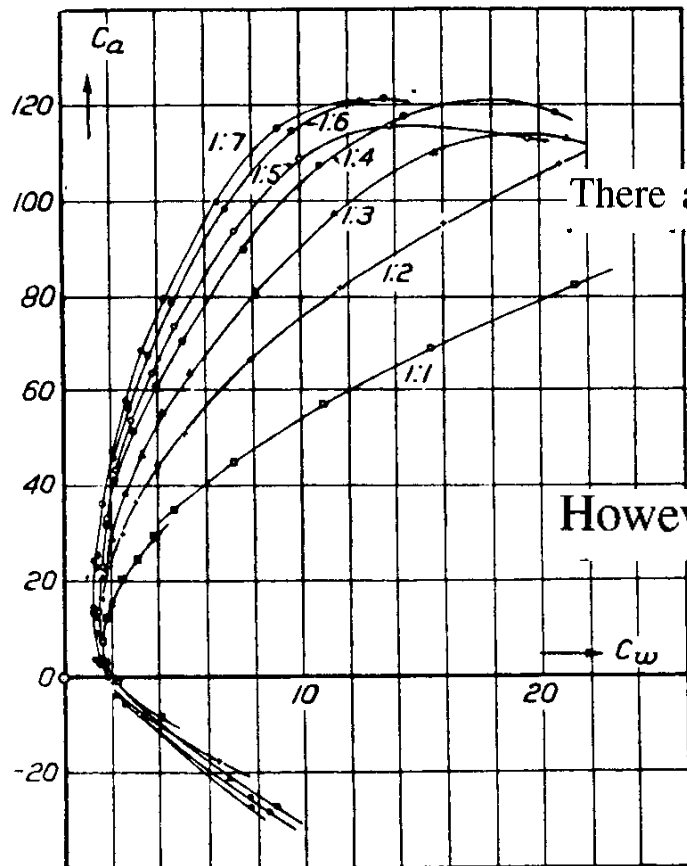


distribution closely approximates the elliptic case. The variation of δ as a function of taper ratio for wings of different aspect ratio is illustrated in Figure 5.18. Such calculations of δ were first performed by the famous English aerodynamicist, Hermann Glauert and published in Reference 18 in the year 1926. Note from Figure 5.18 that a tapered wing can be designed with an induced drag coefficient reasonably close to the minimum value. In addition, tapered wings with straight leading and trailing



EFFECT OF ASPECT RATIO

Note that AR, which typically varies from 6 to 22 for standard subsonic airplanes and sailplanes, has a much stronger effect on $C_{D,i}$ than the value of δ , which from Figure 5.18 varies only by about 10 percent over the practical range of taper ratio. Hence, the primary design factor for minimizing induced drag is not the closeness to an elliptical lift distribution, but rather, the ability to make the aspect ratio as large



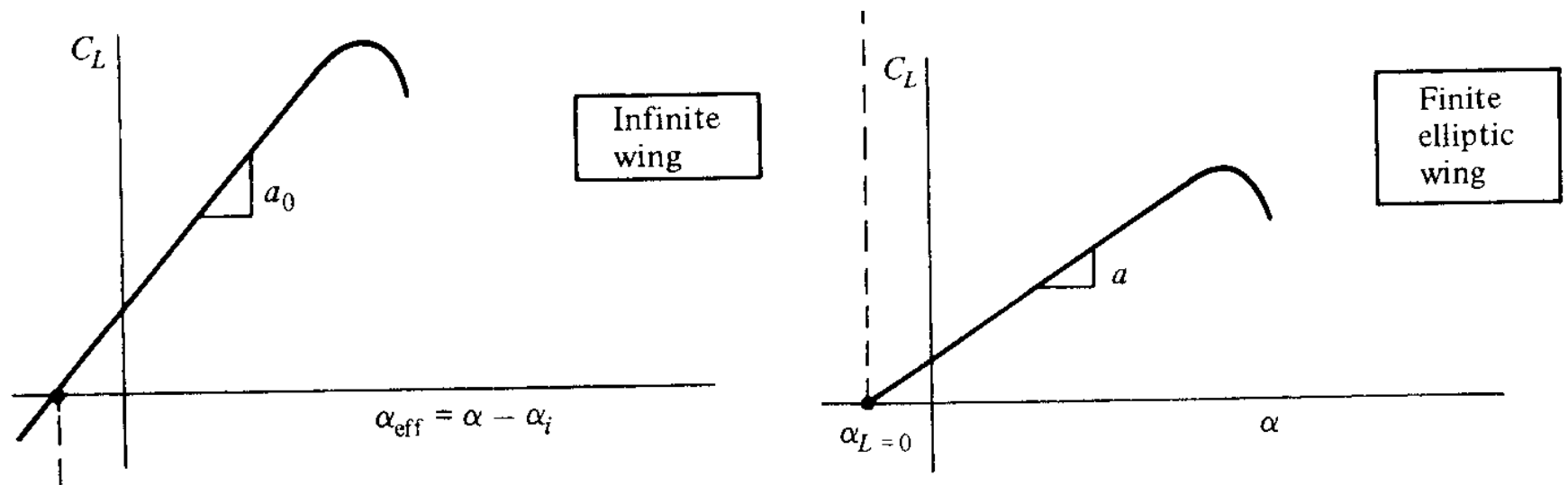
$$C_D = c_d + \frac{C_L^2}{\pi e AR}$$

There are two primary differences between airfoil and finite-wing properties.

a finite wing generates induced drag.

However, a second major difference appears in the lift slope.

slope for an airfoil was defined as $a_0 \equiv dc_l/d\alpha$. Let us denote the lift slope for a finite wing as $a \equiv dC_L/d\alpha$. When the lift slope of a finite wing is compared with that of its airfoil section, we find that $a < a_0$. To see this more clearly, return to Figure 5.4, which illustrates the influence of downwash on the flow over a local airfoil section of a finite wing. Note that although the geometric angle of attack of the finite wing is α , the airfoil section effectively senses a smaller angle of attack, namely, α_{eff} , where $\alpha_{\text{eff}} = \alpha - \alpha_i$. For the time being, consider an elliptic wing with no twist; hence, α_i and α_{eff} are both constant along the span. Moreover, c_l is also constant along the span, and therefore $C_L = c_l$. Assume that we plot C_L for the finite wing versus α_{eff} , as shown at the top of Figure 5.21. Because we are using α_{eff} the lift slope corresponds



to that for an infinite wing a_0 . However, in real life, our naked eyes cannot see α_{eff} ; instead, what we actually observe is a finite wing with a certain angle between the chord line and the relative wind; that is, in practice, we always observe the geometric angle of attack α . Hence, C_L for a finite wing is generally given as a function of α , as sketched at the bottom of Figure 5.21. Since $\alpha > \alpha_{\text{eff}}$, the bottom abscissa is stretched, and hence the bottom lift curve is less inclined; it has a slope equal to a , and Figure 5.21 clearly shows that $a < a_0$. The effect of a finite wing is to *reduce* the lift slope. Also, recall that at zero lift, there are no induced effects; i.e., $\alpha_i = C_{D,i} = 0$. Thus, when $C_L = 0$, $\alpha = \alpha_{\text{eff}}$. As a result, $\alpha_{L=0}$ is the same for the finite and the infinite wings, as shown in Figure 5.21.

$$\frac{dC_L}{d(\alpha - \alpha_i)} = a_0 \quad C_L = a_0(\alpha - \alpha_i) + \text{const} \quad C_L = a_0 \left(\alpha - \frac{C_L}{\pi AR} \right) + \text{const}$$

$$\frac{dC_L}{d\alpha} = a = \frac{a_0}{1 + a_0/\pi AR}$$

Equation (5.69) gives the desired relation between a_0 and a for an elliptic finite wing. For a finite wing of general planform, Equation (5.69) is slightly modified, as given below:

$$a = \frac{a_0}{1 + (a_0/\pi AR)(1 + \tau)} \quad \textbf{[5.70]}$$

In Equation (5.70), τ is a function of the Fourier coefficients A_n .

Values of τ typically range between 0.05 and 0.25

Of most importance in Equations (5.69) and (5.70) is the aspect-ratio variation. Note that for low-AR wings, a substantial difference can exist between a_0 and a . However, as $AR \rightarrow \infty$, $a \rightarrow a_0$. The effect of aspect ratio on the lift curve is dramatically shown in Figure 5.22, which gives classic data obtained on rectangular wings by Prandtl in 1915. Note the reduction in $dC_L/d\alpha$ as AR is reduced. Moreover, using the equations obtained above, Prandtl scaled the data in Figure 5.22 to correspond to an aspect ratio of 5; his results collapsed to essentially the same curve, as shown in

the wing, along with a vortex sheet which trails downstream. This trailing-vortex sheet is the instrument that induces downwash at the lifting line. At first thought, you might consider this model to be somewhat abstract—a mathematical convenience that somehow produces surprisingly useful results. However, to the contrary, the model shown in Figure 5.13 has real physical significance. To see this more clearly, return to Figure 5.1. Note that in the three-dimensional flow over a finite wing, the streamlines leaving the trailing edge from the top and bottom surfaces are in different directions; that is, there is a discontinuity in the tangential velocity at the trailing edge. We know from Chapter 4 that a discontinuous change in tangential velocity is theoretically allowed across a vortex sheet. In real life, such discontinuities do not exist; rather, the different velocities at the trailing edge generate a thin region of large velocity gradients—a thin region of shear flow with very large vorticity. Hence, a sheet of vorticity actually trails downstream from the trailing edge of a finite wing. This sheet

tends to roll up at the edges and helps to form the wing-tip vortices sketched in Figure 5.2. Thus, Prandtl's lifting-line model with its trailing-vortex sheet is physically consistent with the actual flow downstream of a finite wing.

Consider a rectangular wing with an aspect ratio of 6, an induced drag factor $\delta = 0.055$, and a zero-lift angle of attack of -2° . At an angle of attack of 3.4° , the induced drag coefficient for this wing is 0.01. Calculate the induced drag coefficient for a similar wing (a rectangular wing with the same airfoil section) at the same angle of attack, but with an aspect ratio of 10. Assume that the induced factors for drag and the lift slope, δ and τ , respectively, are equal to each other (i.e., $\delta = \tau$). Also, for $AR = 10$, $\delta = 0.105$.

Solution

We must recall that although the angle of attack is the same for the two cases compared here ($AR = 6$ and 10), the value of C_L is different because of the aspect-ratio effect on the lift slope. First, let us calculate C_L for the wing with aspect ratio 6. From Equation (5.61),

$$C_L^2 = \frac{\pi AR C_{D,i}}{1 + \delta} = \frac{\pi(6)(0.01)}{1 + 0.055} = 0.1787$$

Hence,

$$C_L = 0.423$$

The lift slope of this wing is therefore

$$\frac{dC_L}{d\alpha} = \frac{0.423}{3.4^\circ - (-2^\circ)} = 0.078/\text{degree} = 4.485/\text{rad}$$

The lift slope for the airfoil (the infinite wing) can be obtained from Equation (5.70):

$$\frac{dC_L}{d\alpha} = a = \frac{a_0}{1 + (a_0/\pi AR)(1 + \tau)}$$

$$4.485 = \frac{a_0}{1 + [(1.055)a_0/\pi(6)]} = \frac{a_0}{1 + 0.056a_0}$$

Solving for a_0 , we find that this yields $a_0 = 5.989/\text{rad}$. Since the second wing (with $AR = 10$) has the same airfoil section, then a_0 is the same. The lift slope of the second wing is given by

$$a = \frac{a_0}{1 + (a_0/\pi AR)(1 + \tau)} = \frac{5.989}{1 + [(5.989)(1.105)/\pi(10)]} = 4.95/\text{rad}$$

$$= 0.086/\text{degree}$$

The lift coefficient for the second wing is therefore

$$C_L = a(\alpha - \alpha_{L=0}) = 0.086[3.4^\circ - (-2^\circ)] = 0.464$$

In turn, the induced drag coefficient is

$$C_{D,i} = \frac{C_L^2}{\pi AR}(1 + \delta) = \frac{(0.464)^2(1.105)}{\pi(10)} = \boxed{0.0076}$$

Induced Drag

$$D_i = q_\infty S C_{D,i} = q_\infty S \frac{C_L^2}{\pi e AR}$$

$$C_L^2 = \left(\frac{L}{q_\infty S} \right)^2 = \left(\frac{W}{q_\infty S} \right)^2$$

$$D_i = q_\infty S \left(\frac{W}{q_\infty S} \right)^2 \frac{1}{\pi e AR} = \frac{1}{\pi e} q_\infty S \left(\frac{W}{q_\infty S} \right)^2 \left(\frac{S}{b^2} \right)$$

$$D_i = \frac{1}{\pi e q_\infty} \left(\frac{W}{b} \right)^2$$

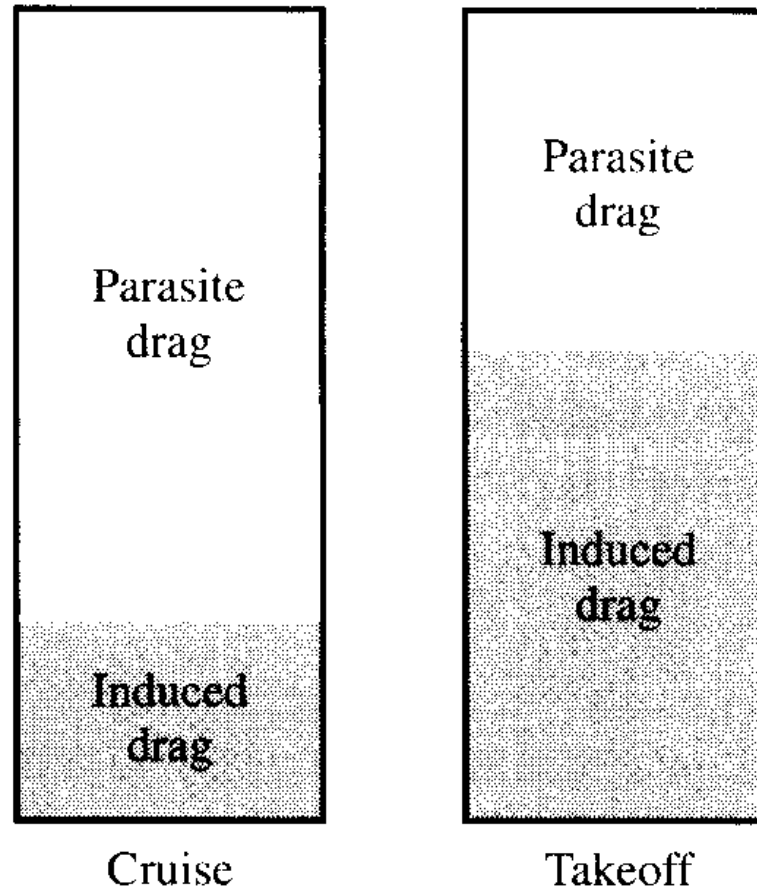
This is a rather revealing result! The induced drag in steady, level flight—the force itself—does not depend explicitly on the aspect ratio, but rather on another design parameter W/b called the *span loading*:

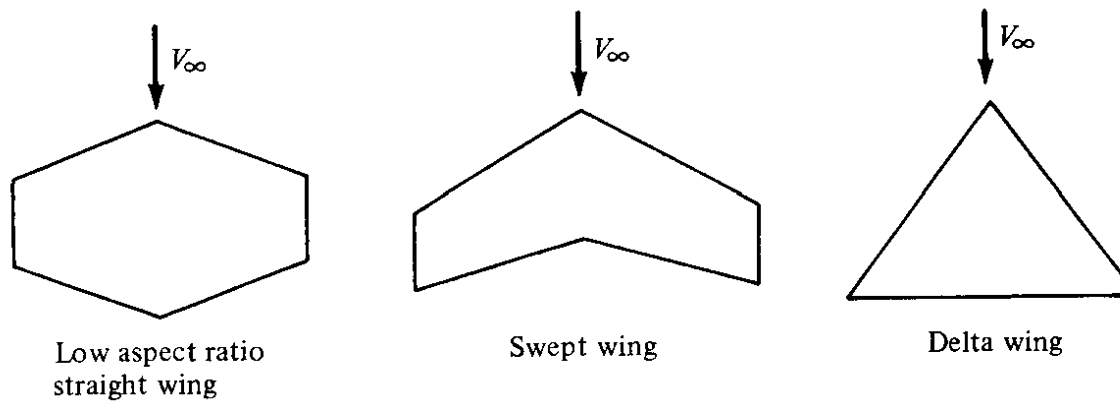
$$\text{Span loading} \equiv \frac{W}{b}$$

From Equation (5.74), we see that the induced drag for a given weight airplane, can be reduced simply by increasing the wingspan b . In so doing, the wing tip vortices (the physical source of induced drag) are simply moved farther away, hence lessening their effect on the rest of the wing and in turn reducing the induced drag. This makes good intuitive sense.

The parasite drag

is the sum of the drag due to skin friction and pressure drag due to flow separation





The lift slope for a high-aspect-ratio straight wing with an elliptical lift distribution is predicted by Prandtl's lifting-line theory and is given by Equation (5.69), repeated below:

$$a = \frac{a_0}{1 + a_0/\pi AR} \quad \text{high-aspect-ratio straight wings} \quad \mathbf{[5.69]}$$

is valid only for high-aspect-ratio straight wings ($AR > 4$, as a rule of thumb)

The German aerodynamicist H. B. Helmbold in 1942 modified Equation (5.69) to obtain the following form applicable to low-aspect-ratio straight wings:

$$a = \frac{a_0}{\sqrt{1 + (a_0/\pi AR)^2} + a_0/(\pi AR)} \quad \text{low-aspect-ratio straight wing} \quad \mathbf{[5.81]}$$

Equation (5.81) is remarkably accurate for wings with $AR < 4$.

For swept wings, Kuchemann (Reference 70) suggests the following modification to Helmbold's equation:

$$a = \frac{a_0 \cos \Lambda}{\sqrt{1 + [(a_0 \cos \Lambda)/(\pi AR)]^2} + (a_0 \cos \Lambda)/(\pi AR)}$$

swept wing **[5.82]**

Swept wings as a means of reducing [wave drag](#)

$$AR \equiv \frac{b^2}{S}$$

$$\text{Taper ratio} \equiv \frac{c_t}{c_r}$$

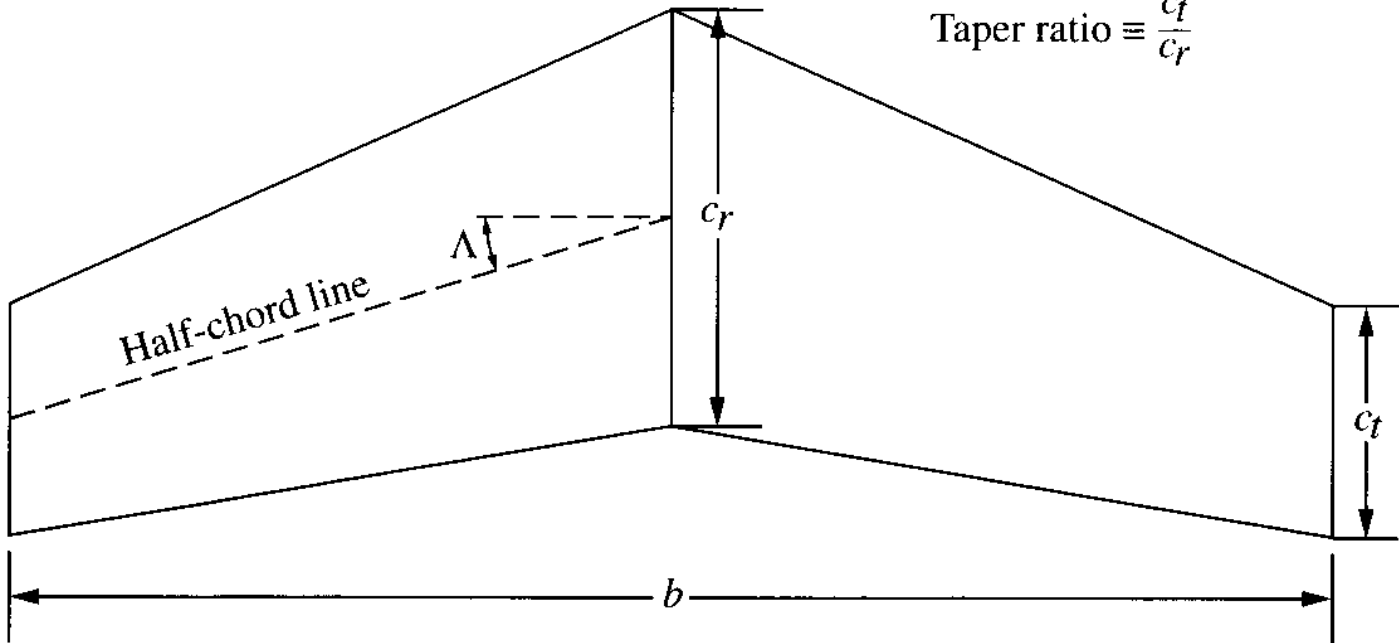
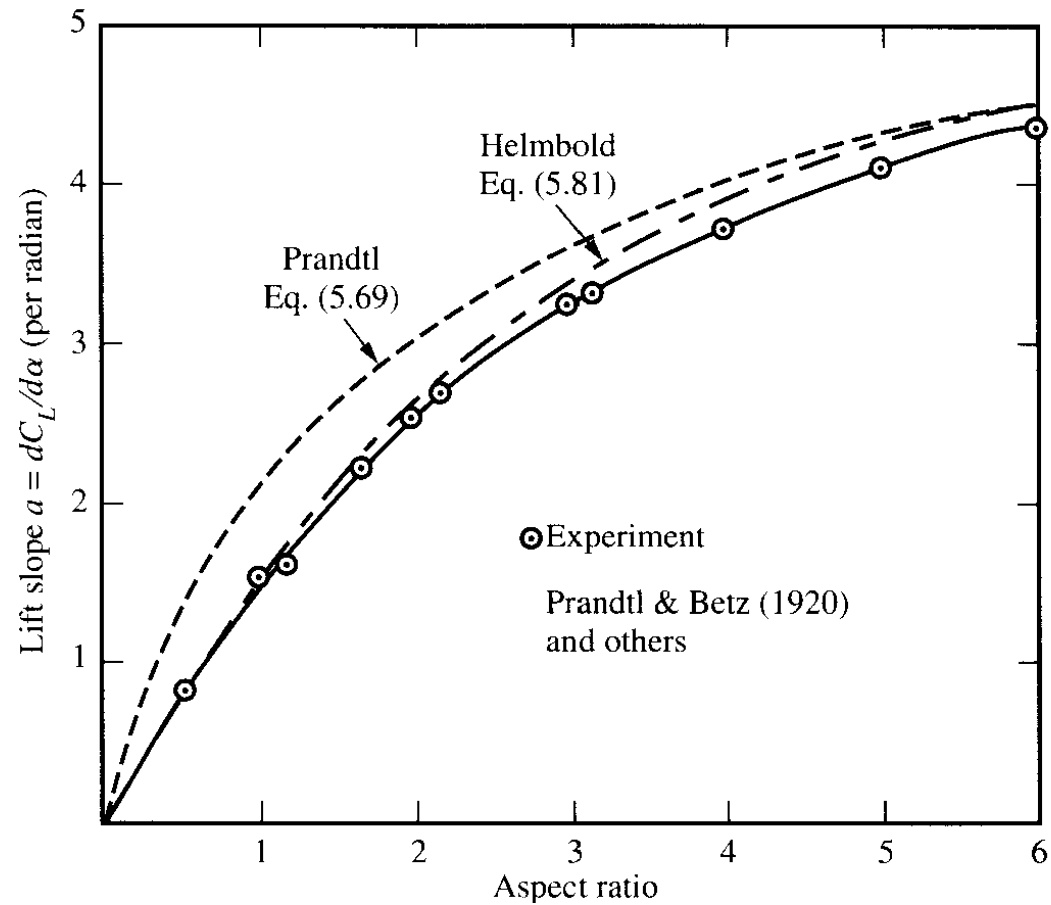
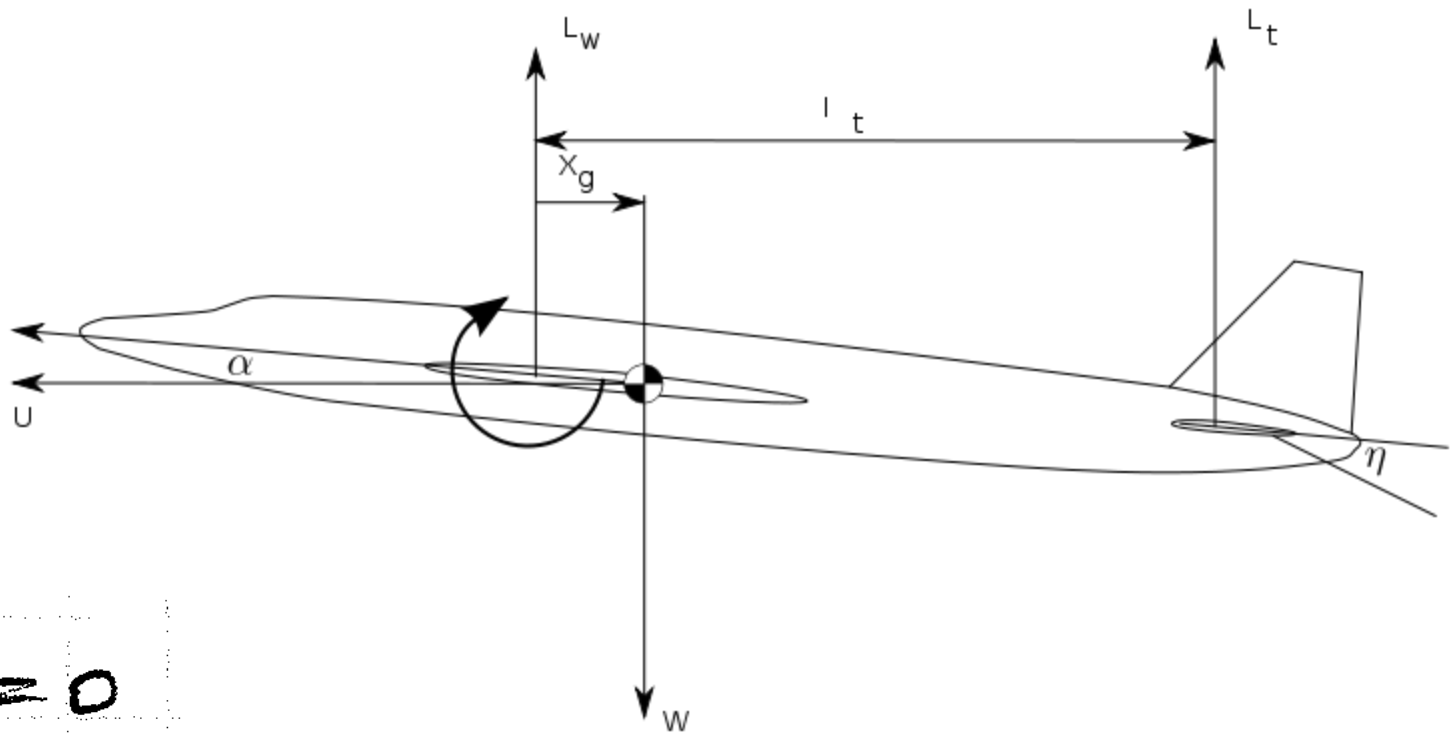


Figure 5.36 Geometry of a swept wing.

Comparison





Aircraft in Level Flight

$$C_{macg} = 0$$

$$M = L_w x_g - (l_t - x_g) L_t \quad \frac{\partial M}{\partial \alpha} = x_g \frac{\partial L_w}{\partial \alpha} - (l_t - x_g) \frac{\partial L_t}{\partial \alpha}$$

$$0 = C_{macw} - C_{Lw} (\hat{x}_w - \hat{x}_{cg}) + C_{macT} \frac{S_T}{S_w} \frac{\bar{c}_T}{\bar{c}_w} - C_{LT} \frac{S_T}{S_w} (\hat{x}_T - \hat{x}_{cg})$$

For symmetric airfoil sections $C_{mac} = 0$

- **Neutral point**
- A mathematical analysis of the longitudinal static stability of a complete aircraft (including horizontal stabilizer) yields the position of center of gravity at which stability is neutral. This position is called the neutral point. (The larger the area of the horizontal stabilizer, and the greater the moment arm of the horizontal stabilizer about the aerodynamic center, the further aft is the neutral point.)
- The static center of gravity margin (c.g. margin) or [static margin](#) is the distance between the center of gravity (or mass) and the neutral point. It is usually quoted as a percentage of the [Mean Aerodynamic Chord](#). The center of gravity must lie ahead of the neutral point for positive stability (negative static margin). If the center of gravity is behind the neutral point, the aircraft is longitudinally unstable (the static margin is positive), and active inputs to the control surfaces are required to maintain stable flight. Ultimately, the position of the center of gravity relative to the neutral point determines the stability, control forces, and controllability of the vehicle.
- For a [tailless aircraft](#) $V_t = 0$, the neutral point coincides with the aerodynamic center, and so for longitudinal static stability the center of gravity must lie ahead of the aerodynamic center.